

Disease Prediction with ICU Mortality



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Agenda

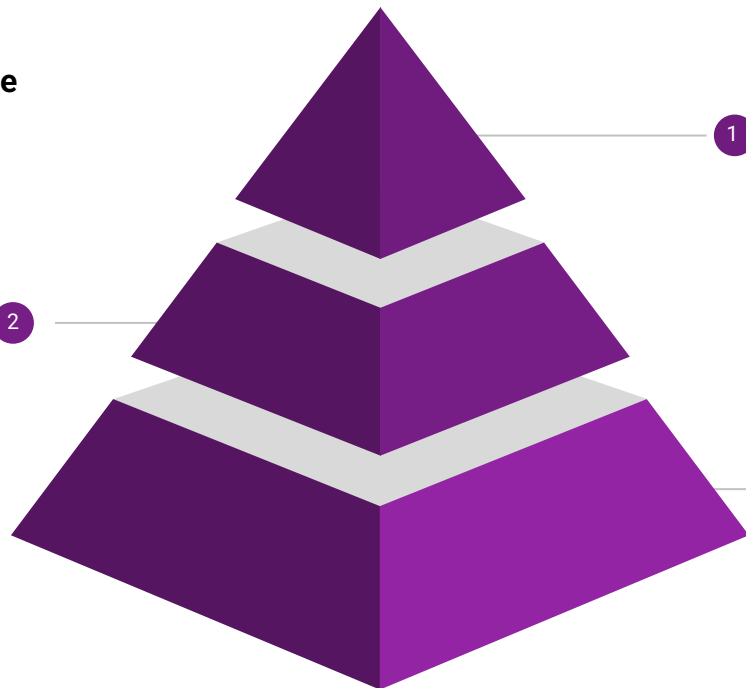
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Introduction

- Healthcare systems generate huge amounts of patient data every day from ICU monitors, laboratory tests, Electronic Health Records (EHR), and clinical reports.
- Early prediction of disease severity and patient mortality is one of the biggest challenges in critical care management.
- Delayed diagnosis or inaccurate risk assessment can increase ICU burden, treatment cost, and mortality rate.
- Machine Learning (ML) provides intelligent techniques to analyze complex medical datasets and identify hidden patterns for accurate prediction.
- This project focuses on **Disease Classification and ICU Mortality Prediction** using ML models trained on clinical and physiological patient data.
- The system aims to classify diseases and predict the survival probability of ICU patients at an early stage.
- Various ML algorithms such as Logistic Regression, Random Forest, Decision Tree, XGBoost, and Deep Learning models can be used for prediction and comparison.
- The project helps doctors and healthcare professionals in making faster and data-driven clinical decisions.

Machine Learning Relevance

- ML models can process large-scale medical datasets efficiently.
- Identifies complex relationships between patient vitals, lab parameters, and outcomes.
- Improves prediction accuracy compared to traditional statistical approaches.
- Enables real-time monitoring and intelligent clinical decision support systems.
- Can continuously improve performance with more healthcare data.
- Provides scalable and automated healthcare analytics solutions.



Healthcare Relevance

- Supports early detection of critical diseases.
- Helps in reducing ICU mortality rate through timely intervention.
- Assists doctors in patient risk stratification and treatment prioritization.
- Improves hospital resource management and ICU bed allocation.
- Enhances quality of patient care using predictive analytics.
- Useful in emergency and critical care environments where rapid decisions are required.

Interdisciplinary Approach

- Combines **Healthcare + Artificial Intelligence + Data Science**.
- Uses medical knowledge for selecting relevant clinical parameters and disease indicators.
- Applies ML algorithms for prediction, classification, and pattern recognition.
- Involves data preprocessing, feature engineering, visualization, and model evaluation.
- Integrates concepts from computer science, biomedical engineering, and healthcare informatics.
- Encourages collaboration between doctors, data scientists, and software engineers for smarter healthcare systems.

Motivation

THE REALITY BEHIND THE NUMBERS

Why This Project is Needed?

THE BASELINE STATISTIC

India reports a

1:834

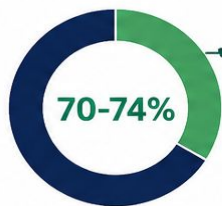
doctor-population ratio



Technically fulfilling the WHO standard (**1:1000**)

THE DISTRIBUTION BOTTLENECK

Skewed distribution of doctors



70%–74%

of doctors work in Urban / Metro areas

Rural & semi-urban areas and govt. hospitals remain severely short-staffed

THE CRITICAL CARE / ICU CRISIS



- **ICU Specialists** are extremely limited.
- One doctor monitors many patients simultaneously.
- Continuous tracking of vitals, labs & devices leads to **high cognitive fatigue** and risk of human error.

ON PAPER (NATIONAL AVERAGE)

- ✓ 1:834 doctor-population ratio
- ✓ Meets WHO standard (**1:1000**)
- ✓ Looks satisfactory

VS

ON GROUND (REALITY)

- ✗ Doctors concentrated in cities
- ✗ Rural & public hospitals face severe shortage
- ✗ ICU workload is extremely challenging
- ✗ Delays in critical decisions affect patient outcomes

Motivation

- High patient load and fewer doctors make continuous monitoring difficult.
- ICU patients require quick monitoring and timely decisions.
- Delays in treatment may increase mortality risk.
- Large healthcare data is difficult to analyze manually.

The motivation behind this project is to assist doctors in handling increasing ICU patient load and enable faster, data-driven medical decisions to reduce mortality risk and eliminating the burden on the healthcare workforce through resource optimization and predictive analytics.

Objective and Problem Statement

Problem Statement

- ICU patients require continuous monitoring and rapid medical decisions.
- Traditional prediction methods may fail to identify high-risk patients at an early stage.
- Large amounts of healthcare data are difficult to analyze manually.
- An intelligent ML-based system is needed for disease classification and ICU mortality prediction to improve patient care and reduce mortality risk.

Objectives

- To classify diseases using patient clinical data.
- To predict ICU mortality using Machine Learning models.
- To analyze healthcare datasets and identify important medical features.
- To compare different ML algorithms for prediction accuracy.
- To support doctors with faster and data-driven decision making.

Literature Review

S. No.	Title	Conclusion	Findings
1	Machine Learning and Early Warning Systems for ICU Mortality Prediction Using MIMIC Data – Johnson A. E. W. (2024)	Machine learning-based early warning systems significantly improve ICU mortality prediction accuracy and help clinicians make faster decisions.	• Logistic Regression – AUROC 0.82 • Random Forest – AUROC 0.87 • XGBoost – AUROC 0.90 (Best Performance)
2	Advanced Mortality Prediction in Adult ICU: Introducing a Deep Learning Approach in Healthcare – Dimitrios Simopoulos (2024)	Deep learning models outperform traditional statistical techniques in ICU mortality prediction tasks.	• Deep Neural Network (DNN) – Accuracy 93% • LSTM – Accuracy 95% (Best Performance)
3	Systematic Reviews of Machine Learning Applications in Healthcare – Katarzyna Kolasa (2023)	Machine learning has become an effective tool for improving clinical decision-making and healthcare efficiency.	• Support Vector Machine (SVM) – Accuracy 82% • Artificial Neural Network (ANN) – Accuracy 88% • Random Forest – Accuracy 90% (Best Performance)

4	Scalable and Accurate Deep Learning for Electronic Health Records – Alvin Rajkomar (2023)	Deep learning can efficiently process large-scale hospital data and improve clinical prediction accuracy.	• Logistic Regression – AUROC 0.85 • Deep Learning EHR Model – AUROC 0.94 (Best Performance)
5	Machine Learning Models Predict Mechanical Ventilation Duration in ARDS Patients – Weng Wei-Hung (2022)	Machine learning models can accurately estimate ventilator duration and improve ICU resource planning.	• Random Forest – AUROC 0.85 • Gradient Boosting – AUROC 0.89 (Best Performance)
6	Deep Learning Enabled Clinical Prediction Systems for Critical Care Applications – Andre Esteva (2022)	Deep learning models provide reliable clinical predictions and support healthcare professionals in emergencies.	• CNN – Accuracy 90% • RNN – Accuracy 92% • Deep Neural Network – Accuracy 94% (Best Performance)
7	Predicting ICU Length of Stay Using Machine Learning – Muhammad Faisal (2021)	Machine learning algorithms successfully estimate ICU stay duration, helping improve hospital resource allocation.	• Decision Tree – Accuracy 84% • Random Forest – Accuracy 86% • XGBoost – Accuracy 88% (Best Performance)

Methodology: Dataset

The dataset used in this project was taken from Kaggle. The dataset contained ICU patient records with different clinical parameters related to patient condition, organ function, laboratory reports, and survival outcome.

RecordID	Age	Gender	Height	Albumin	ALP	ALT	AST	Bilirubin	BUN	Cholesterol	Creatinine	DiasA
140574	79	1	172.7						25.25		1.66	
140589	70	1	177.8						19.75		0.8	
140202	56	0	160	2.6	69	25	19	0.3	10		0.33	
140165	72	0							63.44		1.74	
139655	72	1	175.3						14.67		0.83	
141086	62	1			42	12	27.5	2.87	28.67		1.1	
139859	71	0	154.9		52	168	202	1.67	11		0.67	
139533	60	1	172.7	2.36	42.8	350.6	422.4	1.82	26.8		1.1	
139976	38	0	167.6		102			0.55	30		1.08	

Fig 02: Original Dataset

The original dataset contained around 4,000+ patient records. However, a large number of records had missing values and incomplete information. Many patient parameters were empty in several records, so the complete dataset could not be used directly for machine learning training.

Problems Faced:

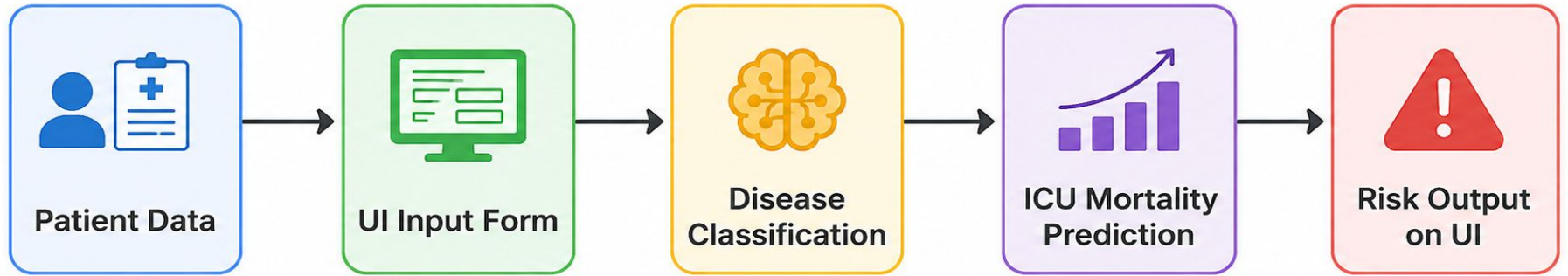
Because of this problem, two important steps were performed during dataset preparation:

- Records having maximum available parameter values were selected for use.
- Some parameters with small missing ranges were completed using Python-based augmentation and imputation techniques by assigning values within acceptable clinical ranges.

1	GCS	Glucose	HCO3	HCT	HR	K	Lactate	Mg	MAP	MechVent	Na	NIDias	ABf	NI	MAP	NI	SysABP	PaCO2	PaO2	pH	Platelets	SaO2	SysABP	Temp	Urine	WBC	Weight	ICUType	completeness
2	13.3	95.25	25	36.85	64.92	3.4	5.8	1.83	79.27	1	142	49.29	76.57	131.14	35.17	242.83	7.43	179.25	98	124.61	36.56	117.09	9.73	71.15	1	38			
3	10.42	166.5	21	34.33	94.79	4.15	2.9	2.05	88.02	1	141	71.33	91.54	131.94	32.33	161.5	7.38	205	99.33	111.18	36.91	38.82	18.03	67	1	38			
4	9.15	147.5	23.4	31.82	79.15	4.16	3.25	1.59	67.27	1	127.5	59.43	73.38	101.29	38.18	121.64	7.38	375.75	96.64	94.7	36.25	27.06	21.05	75	1	38			
5	10	142.5	26	41.24	99.22	4.02	2	2.32	80.45	1	132.75	52	72.03	112.08	38	135.57	7.4	248.6	95.75	112	37.23	150.17	15.38	79.9	1	38			
6	9.89	155.83	17.33	28.18	102.93	5.17	1.8	2.56	76.54	1	139.5	49.2	65.73	98.8	33.73	80.2	7.36	288	94.92	105.56	37.81	44.21	14.48	107.4	1	38			
7	11.45	146	26.6	31.48	95.54	4.37	1.45	2.68	74.54	1	139.2	62.39	79.76	114.5	45.67	180.22	7.39	98	97.38	114.77	36.94	77.33	8.6	71.57	2	38			
8	9.31	136.33	19.67	34.38	76.04	3.46	2.7	1.9	78.45	1	138.33	47.07	66.54	105.46	33.77	139.77	7.41	263.67	98.25	124.38	37.87	65.67	18.2	71.01	2	38			
9	9.96	152.33	23.67	25.2	71.17	4.83	1.6	1.4	64.95	1	139	29.5	51.67	101.75	46.57	132.29	7.34	65.33	98	108.79	36.52	74.49	15.3	107.76	4	38			
10	10.82	91	18.67	29.68	77.16	4.07	0.97	1.93	62.05	1	133.33	47.03	62.07	92.16	32.78	148.44	7.32	163.25	98	97.88	36.72	37.5	15.75	71.68	1	38			
11	5.91	237.8	22.6	34.2	106.77	4.04	1.53	1.78	69.61	1	135.8	53.18	65.27	89.45	41.22	84.33	7.35	258	95.67	97.89	37.54	24.4	13.54	100	3	38			
12	14.8	141.43	19.57	31.14	61.57	5.1	1.3	2.24	68.87	1	132	50.03	61.5	99.58	44.83	135.83	7.29	428.2	97.5	103.85	36.67	3.45	13.42	108.61	1	38			
13	9.28	135.67	20	36.67	88.57	4.03	2.8	2.02	89.12	1	136.67	57.84	72.04	100.42	34.8	110.6	7.39	219.5	94.25	112.36	38.57	38.59	7.5	67.6	1	38			

Fig 02: Snapshot of Augmented Data This helped in improving dataset usability and reducing unnecessary data loss.

Workflow



Patient Data

- The dataset originally contained many clinical parameters, but not all parameters were used for model training. Using too many unnecessary features can increase noise and reduce prediction quality. So, feature selection was performed to identify clinically useful parameters for ICU prediction tasks.
- Initially, around 38 important clinical parameters were selected from the dataset because they were more relevant for ICU monitoring and mortality prediction.
- However, not all 38 parameters were used together for every disease classification task. This helped in reducing unnecessary features during training and improved classification performance.

All parameters included in the training:

Category	Features
Demographic Features	Age, Weight
Hemodynamic Features	MAP, HR
Respiratory Features	PaO2, FiO2, MechVent
Blood & Metabolic Features	Lactate, Glucose, pH, HCO3, WBC, HCT, Platelets
Kidney Function Features	BUN, Creatinine, Urine
Liver Function Features	Albumin
Other Clinical Features	GCS, SOFA

UI Input Form

ICU Mortality Predictor

ICU Mortality Predictor

SOFA		Age	
GCS		BUN	
Creatinine		Urine	
Lactate		pH	
HCO3		MAP	
HR		PaO2	
FiO2		MechVent	
WBC		HCT	
Platelets		Albumin	
Glucose		Weight	

Prediction: --

Probability: --

Low Risk Medium High Risk

Conditions: --

Predict

Refresh

Disease Classification

To categorize the clinical condition of patients based on their physiological and laboratory parameters, a rule-based expert system was implemented. The classification logic evaluates key biomarkers extracted from the patient's data to identify critical syndromic states, including shock, organ failures, and infections.

Target Condition	Primary Parameter	Threshold	Secondary Parameter	Threshold	Logical Operator
Shock	Lactate	>2	MAP	<65	AND
Kidney Failure	Creatinine	>1.5	BUN	>25	AND
Neurological Issue	GCS	<8	-	-	-
Respiratory Failure	PaO2	<60	FiO2	>0.6	OR
Infection	WBC	>12,000	-	-	-

Visualization

Diseases

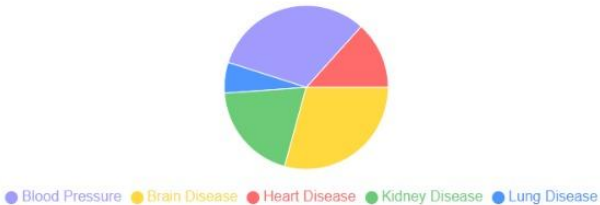
- Heart Disease
- Blood Pressure
- Lung Disease
- Kidney Disease
- Brain Disease

Disease Dashboard

Donut Chart



Pie Chart



Patient Vitals Matrix

Top 10 Patients

S.No.	Patient ID	Heart Rate	BP Systolic	BP Diastolic	MAP	Resp. Rate	BUN	GCS	Temp (°C)
1	132539	73	147	65	92.33	19	13	15	35.1
2	132543	79	134	63	86.67	19	23	15	36.3
3	132547	78	129	89	102.3	-	15	7	35.8
4	132548	73	187	88	121	14	32	15	36.3
5	132551	111	100	51	67.33	-	81	15	38



Reports

ICU Type Report

Mechanical Ventilation
Report

Mortality Report

Number of Beds Report

Length of Stay Report

Gender and Age
Distribution Report

Survival Analysis Report

SAPS & SOFA Report

Future HR Patient Report

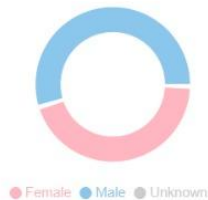
Diabetic Patient Report

Patient Individual Report

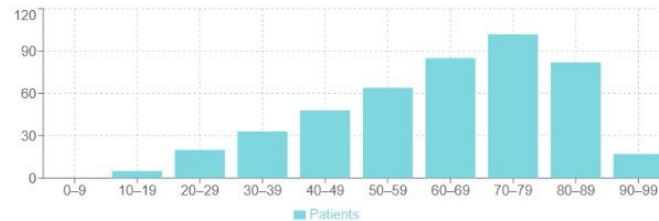
Comparison Patient ID
Report

Gender and Age Distribution Report

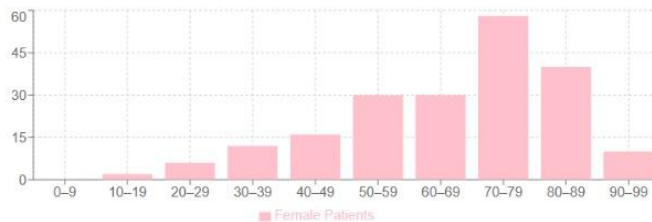
Gender Distribution



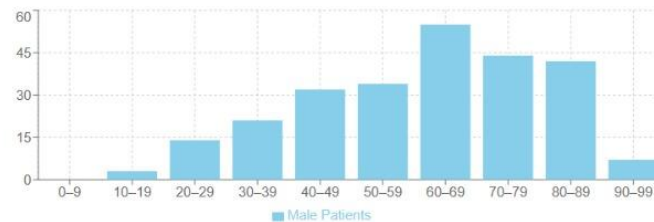
Overall Age Distribution



Female Age Distribution



Male Age Distribution



ICU Mortality Prediction

- The mortality prediction part of the system was made for ICU patients using machine learning. The system checks different patient parameters and predicts whether the patient's condition is more towards survival or high mortality risk. The prediction is based on ICU patient data and important clinical values.
- Some parameters related to the **SOFA score** were also considered in this project because the SOFA is commonly used in ICUs for checking organ condition and patient severity. Different body conditions like respiratory condition, kidney function, neurological condition, and blood pressure-related parameters, were included in the prediction process.
- The system checks these parameters together and then gives mortality prediction results. The output is shown as a survival or non-survival prediction based on the patient data entered into the system. This may help in identifying **high-risk patients** earlier during monitoring.
- The prediction system can also help in situations where many ICU patients need continuous observation together. It may help doctors and hospital staff in checking patient condition faster and deciding which patients may need more attention or monitoring.
- The system is only designed as a support tool for ICU monitoring work. Final treatment decisions and medical judgment should always be made by doctors and healthcare professionals only.

Risk Output on UI

ICU Mortality Predictor

ICU Mortality Predictor

SOFA	1	Age	79
GCS	13.3	BUN	11.8
Creatinine	0.62	Urine	117.09
Lactate	5.8	pH	7.43
HCO3	25	MAP	79.27
HR	64.92	PaO2	242.83
FIO2	0.68	MechVent	1
WBC	9.73	HCT	36.85
Platelets	179.25	Albumin	3.7
Glucose	95.5	Weight	71.15

Prediction: NO (Patient will SURVIVE)

Probability: 4.0%



Low Risk Medium High Risk

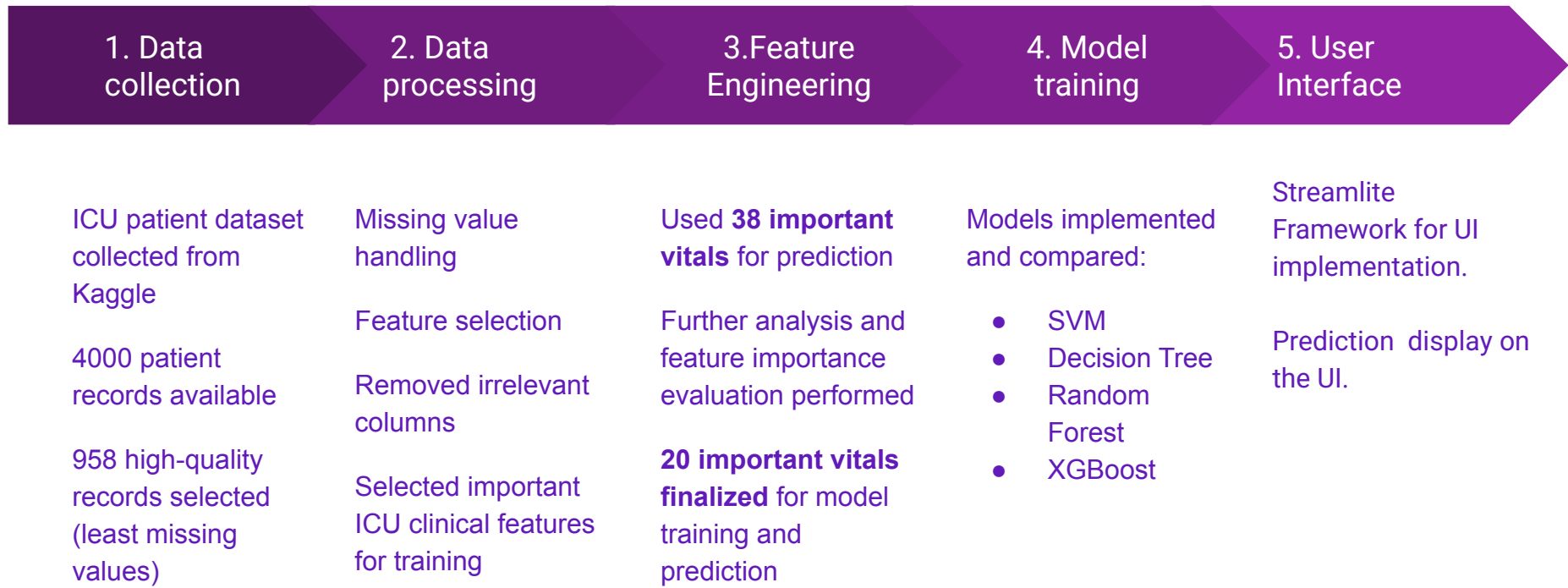
Conditions:

- Respiratory Failure

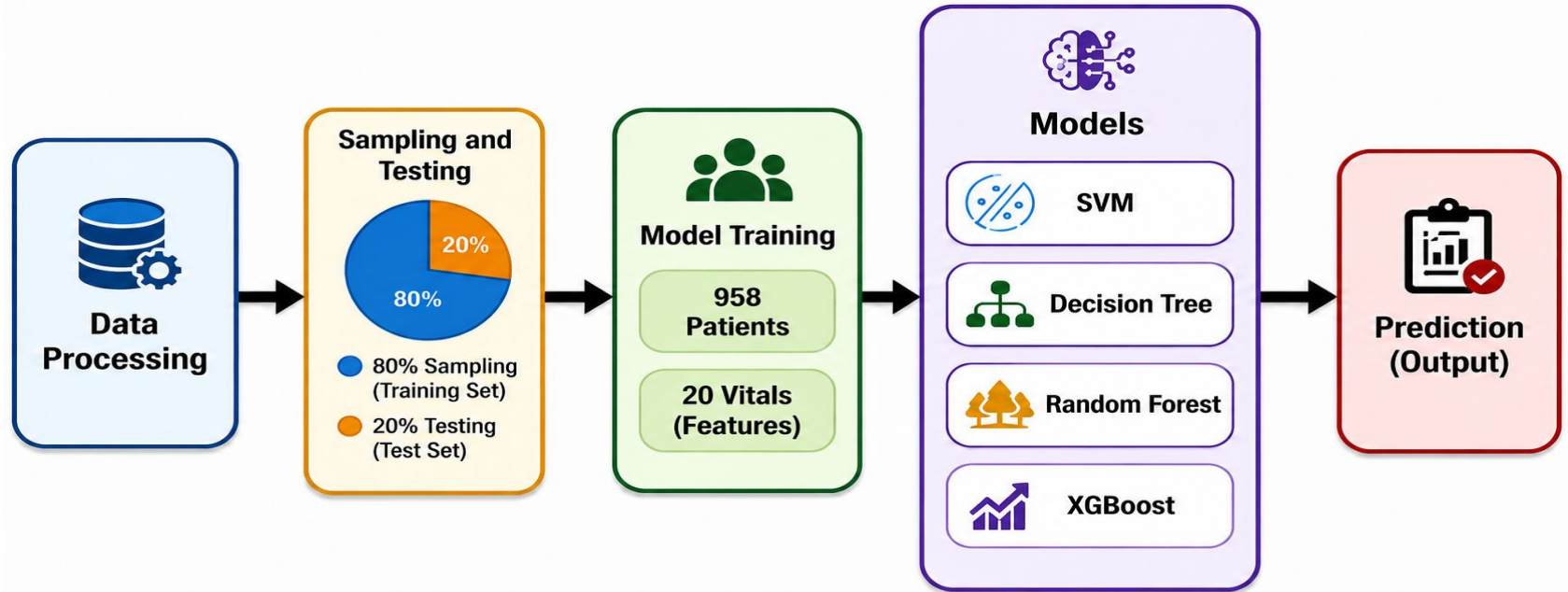
Predict

Refresh

Implementation



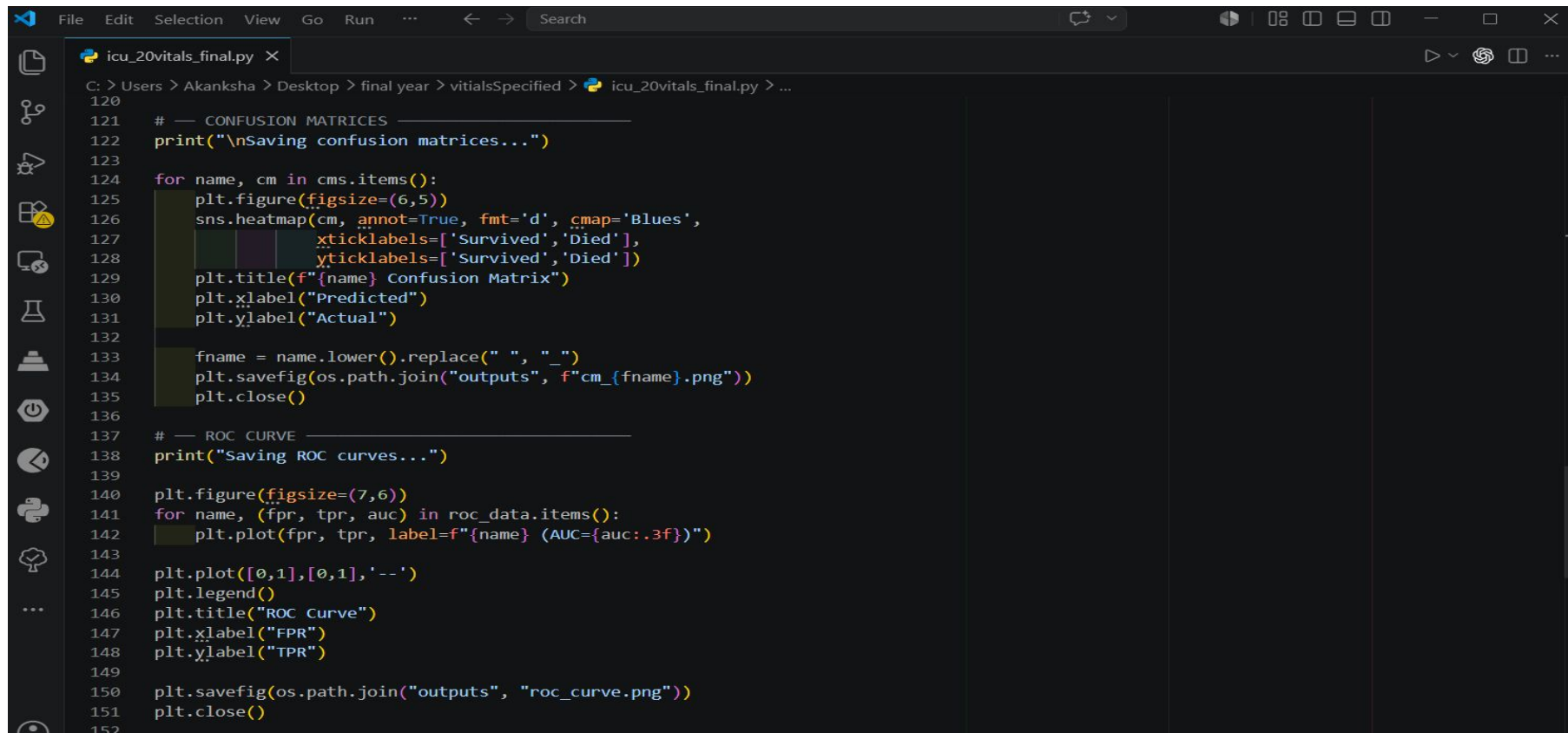
Implementation



Code

```
File Edit Selection View Go Run ... Search
icu_20vitals_final.py X
C: > Users > Akanksha > Desktop > final year > vitalsSpecified > icu_20vitals_final.py
20 # Create output folder
21 os.makedirs("outputs", exist_ok=True)
22
23 # — TOP 20 FEATURES —————
24 TOP_20 = [
25     'SOFA', 'Age', 'GCS', 'BUN', 'Creatinine', 'Urine', 'Lactate',
26     'pH', 'HCO3', 'MAP', 'HR', 'PaO2', 'FiO2', 'MechVent',
27     'WBC', 'HCT', 'Platelets', 'Albumin', 'Glucose', 'Weight'
28 ]
29 TARGET = 'In-hospital_death'
30
31 # — LOAD DATA —————
32 df = pd.read_csv("merged_output.csv")
33
34 df.drop(columns=['TropI', 'TropT'], inplace=True)
35 high_null = [c for c in df.columns if df[c].isnull().mean() > 0.85]
36 df.drop(columns=high_null, inplace=True)
37
38 drop_cols = ['RecordID', 'Survival', 'Length_of_stay', 'completeness']
39 df.drop(columns=[c for c in drop_cols if c in df.columns], inplace=True)
40
41 df.fillna(df.median(numeric_only=True), inplace=True)
42 df.fillna(0, inplace=True)
43
44 X = df[TOP_20]
45 y = df[TARGET]
46
47 print(f"Dataset: {X.shape[0]} patients | {X.shape[1]} vitals")
48 print(f"Class split → Survived: {(y==0).sum()} | Died: {(y==1).sum()}")
49
50 # — TRAIN TEST SPLIT —————
51 X_train, X_test, y_train, y_test = train_test_split(
52     X, y, test_size=0.2, random_state=42, stratify=y)
53
54 # — MODEL TRAINING —————
```

Code

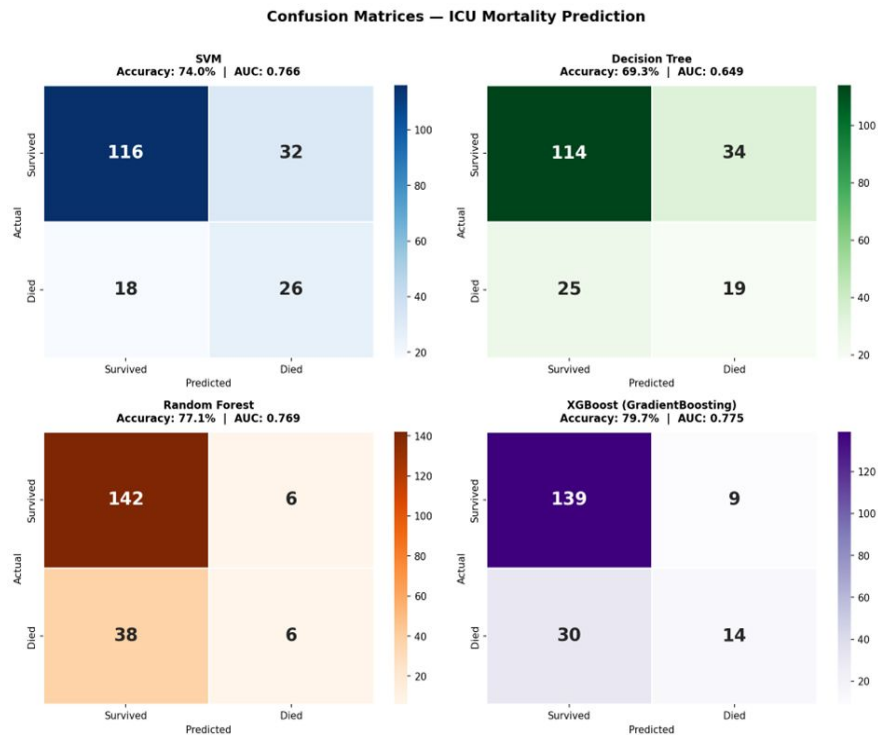


```
File Edit Selection View Go Run ... Search
icu_20vitals_final.py X
C:\Users\Akansha\Desktop>final year>vitalsSpecified>icu_20vitals_final.py> ...
120
121 # — CONFUSION MATRICES —————
122 print("\nSaving confusion matrices...")
123
124 for name, cm in cms.items():
125     plt.figure(figsize=(6,5))
126     sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
127                 xticklabels=['Survived', 'Died'],
128                 yticklabels=['Survived', 'Died'])
129     plt.title(f"{name} Confusion Matrix")
130     plt.xlabel("Predicted")
131     plt.ylabel("Actual")
132
133     fname = name.lower().replace(" ", "_")
134     plt.savefig(os.path.join("outputs", f"cm_{fname}.png"))
135     plt.close()
136
137 # — ROC CURVE —————
138 print("Saving ROC curves...")
139
140 plt.figure(figsize=(7,6))
141 for name, (fpr, tpr, auc) in roc_data.items():
142     plt.plot(fpr, tpr, label=f"{name} (AUC={auc:.3f})")
143
144     plt.plot([0,1],[0,1], '--')
145     plt.legend()
146     plt.title("ROC Curve")
147     plt.xlabel("FPR")
148     plt.ylabel("TPR")
149
150     plt.savefig(os.path.join("outputs", "roc_curve.png"))
151     plt.close()
152
```

Result

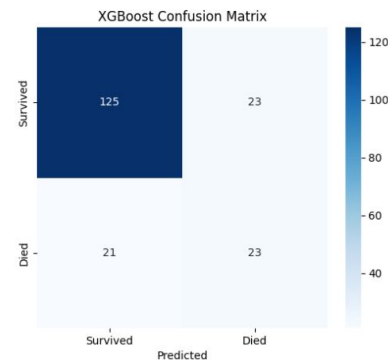
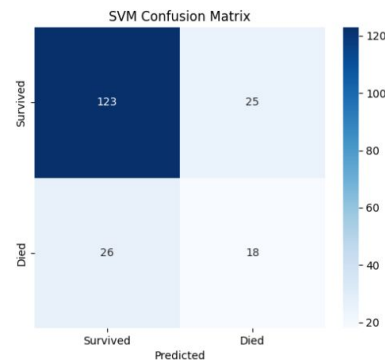
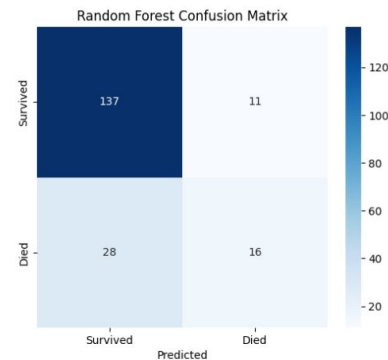
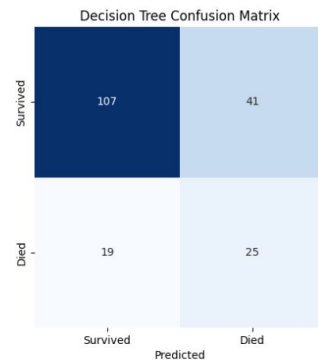
Confusion matrix when we are using 38 parameters

Models (over 38 vitals)	Accuracy	AUC-ROC	True Positive (Deaths Caught)	False Negative (Deaths Missed)
SVM	74.0%	0.766	26	18
Decision Tree	69.3%	0.649	19	25
Random Forest	77.1%	0.769	6	38
XGBoost	79.7%	0.775	14	30



Confusion matrix when we are using 20 parameters

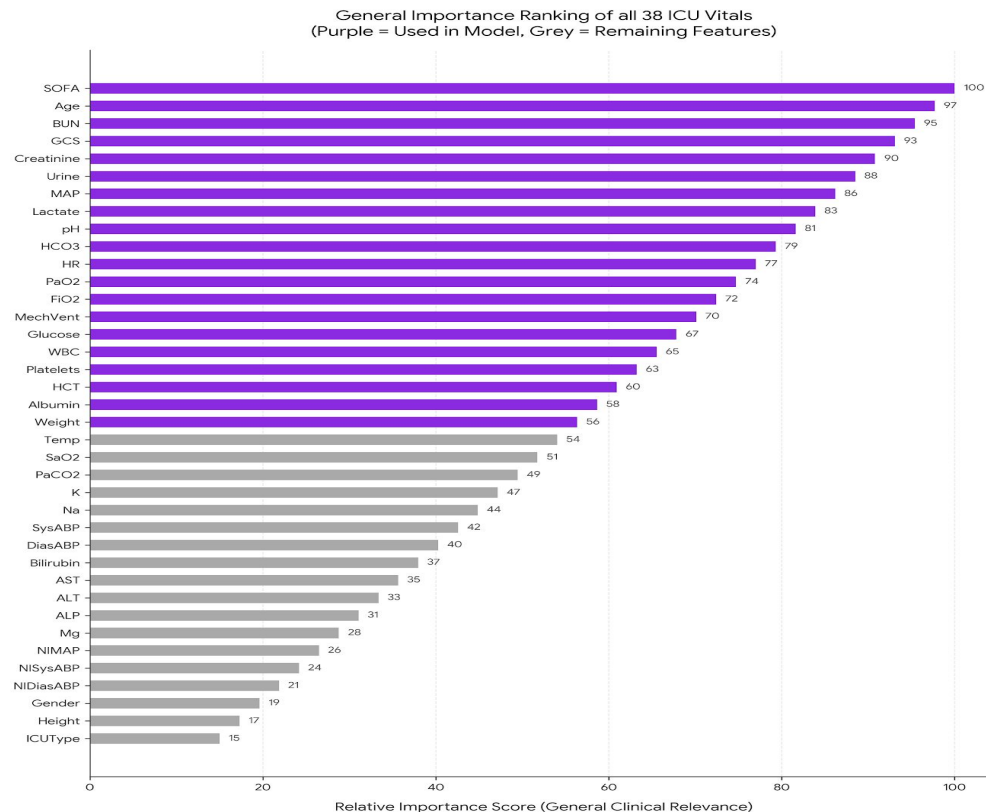
Models (over 20 vitals)	Accuracy	AUC-ROC	True Positive (Deaths Caught)	False Negative (Deaths Missed)
SVM	73.4%	0.760	18	26
Decision Tree	68.8%	0.620	25	19
Random Forest	79.7%	0.782	16	28
XGBoost	76.6%	0.767	23	21



Why 20 Vitals Instead of 38?

- Missing Value Reduction
- Noise Reduction & Better Generalization
- Faster & Efficient Prediction
- Selection of High-Impact Clinical Features

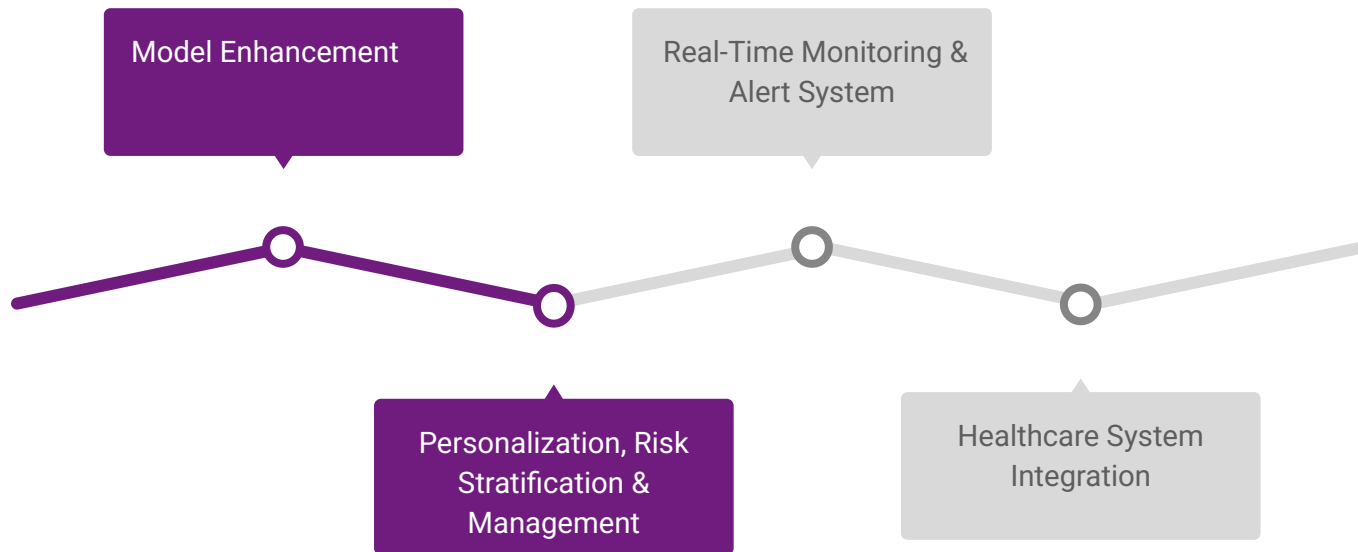
20 highly relevant vitals were selected to improve prediction accuracy, reduce complexity, and support efficient ICU mortality prediction.



Conclusion

- This project demonstrates the use of Machine Learning for disease classification and ICU mortality prediction using clinical patient data.
- Different ML algorithms were implemented and compared for prediction performance.
- Among all models, **Random Forest achieved the best accuracy** and provided more reliable prediction results.
- The system can help identify high-risk ICU patients at an early stage and support faster clinical decision-making.
- Early prediction and intelligent analysis can improve patient care, reduce mortality risk, and optimize ICU management.
- The project highlights the importance of combining healthcare and artificial intelligence for advanced predictive healthcare systems.

Future scope



References

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Thank You